Karan Shah

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Education

Stony Brook University — *Stony Brook, NY*

MS in Computer Science | GPA : 3.64/4 — *Aug. 2019 - **Exp. Dec. 2020***

Coursework: Big Data Analytics, Theory of Database Systems, Machine Learning, Computer Vision, Data Science, Probability and Statistics

Gujarat Technological University — *Ahmedabad, India*

B.E. in Computer Engineering — *Aug. 2014 - May. 2018*

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Skills

IDEs	Visual Studio, NetBeans, Jupyter Notebook, Eclipse, IntelliJ IDEA, Vim, Sublime
Programming	**Python**, Javascript, JAVA, C/C++, Golang
Tools/Technologies	**Matlab**, AWS, MapReduce, Spark, Hadoop, Git, SQL, NoSQL, Linux, Docker, Jenkins, Kubernetes
Frameworks/Libraries	**PyTorch, Tensorflow, Keras, Scikit-Learn**, **OpenCV**, Numpy, Pandas, NodeJS, Django, Flask, Angular, ElectronJS

Work Experience

Stony Brook University — *Stony Brook, NY*

Graduate Teaching Assistant — *January 2020 - May 2020*

- Assisted Prof. Dimitris Samaras in teaching Course CSE327-01: **Computer Vision**
- Managed assignment formulations and assessments, and resolve queries of students enrolled in the course

Simform — *Ahmedabad, India*

Software Engineer — *June 2018 - April 2019*

- Developed and automated ETL jobs to work on a huge realtime data using AWS services (Glue, Lambda, S3) and Spark
- Developed an end to end Video Streaming system; able to upload, scan and host the content on the cloud, and stream on cross-platform desktop application over globally separated clients; using AWS services (Lambda, S3, Cloudfront, MQTT), NodeJS, and Python
- Developed software system to semi-automate Company's marketing work with Chrome Extension and REST API Web backend using Python and NodeJS, efficient in reducing $\sim$ **40 %** of Marketing team's manual work

ISRO - Indian Space Research Organization — *Ahmedabad, India*

Research Intern, Machine Learning — *July 2017 - May 2018*

- Designed generalized **CNN** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for data preparation/processing of geographic remote sensing data, visualization, real-time training performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Projects

Pansharpening Using Convolutional Neural Networks (Python, Keras) — *Jan. 2019 - July 2019*

- Fine-tuned a Convolutional Neural Network to super-resolve Multispectral Satelite Imagery to Panchromatic Resolution
- Delivered high performance in terms of reduced resolution quantitative assessment and visual inspection
- Achieved **better results** compared to popular methods used for this purpose in the industry

Image Denoising using Deep Neural Networks (Python, Keras) — *Oct. 2018 - Dec. 2018*

- Designed a Deep Neural Network for Image Denoising and trained-performed on Overhead Images having Gaussian Noise
- Achieved computational efficiency and ability to generalize well despite limited training data
- Used Worldview Imagery samples provided by DigitalGlobe, achieving **90%** accuracy

Video Action Classification and Recognition (Python, PyTorch) — *Nov. 2019 - Dec. 2019*

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (inclass 3rd rank) on image test data, and **84%** (inclass 4th rank) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

TV Series Scene Classification (Matlab) — *Oct. 2019 - Nov. 2019*

- Trained SVMs with different kernels and used them to classify scenes of a TV series, achieved **84.5%** categorization accuracy on test data
- Used Bag-of-Word representation, LibSVM, and implemented K-means clustering

Ultrasound Nerve Segmentation (Python, Tensorflow, Keras) — *Nov. 2017 - Jan. 2018*

- Designed and trained CNN based on U-net to segment a collection of Nerves in Ultrasound Images taken from a Kaggle competition
- Used pre-trained weights of VGG model in the Encoder of the network (transfer learning), gained **68 %** Test Accuracy (Top 100)

Publication

Band-wise Independent Pansharpening Using Convolutional Neural Networks with Shared Weights

Pankaj Bodani, Karan Shah, Shashikant Sharma

- Submitted to **IEEE** Geoscience and Remote Sensing Letters